Univerzita Komenského v Bratislave
Fakulta matematiky, fyziky a informatiky

IoT platforma GLIDER

Diplomová práca

2026
Nikolas Gubančok

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IoT platforma GLIDER

Diplomová práca

Študijný program:	Aplikovaná informatika
Študijný odbor:	Informatika
Školiace pracovisko:	Katedra aplikovanej informatiky
Školiteľ:	RNDr. Jozef Šiška, PhD.

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THESIS ASSIGNMENT

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Study programme:	Applied Computer Science (Single degree study, master II. deg., full time form)
Field of Study:	Computer Science
Type of Thesis:	Diploma Thesis
Language of Thesis:	English
Secondary language:	Slovak

Title: GLIDER IoT platform

Annotation: IoT platforms serve as hubs that control and collect data from IoT devices. Devices connect to them through various technologies. As some of the technologies become unavailable, such as 2G mobile networks, a lot of systems need to be migrated to newer technologies. Providing a drop in replacement using a newer technology, such as LTE-M, becomes important.
An important factor in IoT platform and device implementation is security (as "the S in IoT"), so care must be taken when designing IoT systems and their communication protocols.
A cloud platform for managing LTE-M devices (GLIDER) was implemented in a previous bachelor thesis aimed at a specific hardware architecture. The next step is to implement the actual software for the devices providing the capabilities of an older system while using a newer communication technology.

Aim: Design and implement software for LTE-M IoT devices that will communicate with the GLIDER platform.

Keywords: IoT, LTE-M

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ZADANIE ZÁVEREČNEJ PRÁCE

Meno a priezvisko študenta:	Bc. Nikolas Gubančok
Študijný program:	aplikovaná informatika (Jednoodborové štúdium, magisterský II. st., denná forma)
Študijný odbor:	informatika
Typ záverečnej práce:	diplomová
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Sekundárny jazyk:	slovenský

Názov: GLIDER IoT platform
IoT platforma GLIDER

Anotácia: Platformy pre internet vecí (IoT) slúžia ako centrálne systémy na ovládanie IoT zariadení a zber dát z nich. Zariadenia sa na platformy pripájajú pomocou rôznych komunikačných technológií. Ako sa niektoré technológie stávajú nedostupnými (2G mobilné siete), mnoho systémov musí byť premigrovaných na novšie technológie. Preto je dôležité zabezpečiť priamočiaru náhradu pomocou novej technológie (napríklad LTE-M).
Dôležitým faktorom pri implementácii IoT systémov je bezpečnosť, preto je dôležité na ňu dbať pri návrhu systémov a komunikačných protokolov.
V rámci predchádzajúcej bakalárskej práce bola implementovaná cloudová platforma na správu LTE-M zariadení (GLIDER), zameraná na konkrétny typ LTE M zariadení. Ďalším krokom je implementácia softvéru pre tieto zariadenia, ktoré budú poskytovať služby ekvivalentné starším 2G zariadeniam používajúc novšiu komunikačnú technológiu.

Cieľ: Navrhnúť a implementovať softvér pre LTE-M IoT zariadenia integrované do platformy GLIDER.

Kľúčové slová: IoT, LTE-M

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Abstrakt

Platformy pre internet vecí (IoT) slúžia ako centrálne systémy na ovládanie IoT zariadení a zber dát z nich. Zariadenia sa na platformy pripájajú pomocou rôznych komunikačných technológií. Ako sa niektoré technológie stávajú nedostupnými (2G mobilné siete), mnoho systémov musí byť premigrovaných na novšie technológie. Preto je dôležité zabezpečiť priamočiaru náhradu pomocou novej technológie (napríklad LTE-M).

Kľúčové slová: IoT, GLIDER, LTE-M, firmvér, nRF9151, MQTT

Abstract

IoT platforms serve as hubs that control and collect data from IoT devices. Devices connect to them through various technologies. As some of the technologies become unavailable, such as 2G mobile networks, a lot of systems need to be migrated to newer technologies. Providing a drop in replacement using a newer technology, such as LTE-M, becomes important.

Keywords: IoT, GLIDER, LTE-M, firmware, nRF9151, MQTT

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Introduction

The rapid growth of the Internet of Things (IoT) has changed the way industrial equipment is monitored and controlled, but a large share of the installed base still relies on legacy 2G GSM/GPRS modems. As mobile network operators progressively shut down their 2G networks, these devices lose their primary transport, and the question of how to migrate them to a successor technology becomes urgent. Long Term Evolution for Machines (LTE-M) is one of the more attractive options here, because it preserves a connection-oriented model similar to GSM while running on infrastructure that operators commit to maintaining for the foreseeable future. Security, which is often listed as the missing "S" in IoT, is just as important as connectivity in this migration, so the design has to take it into account from the start [3].

The Hardwario GLIDER platform was introduced in our previous bachelor's thesis as a cloud service for managing such cellular IoT devices. The work covered the backend, the frontend web application, and a virtual-device emulator that stood in for real hardware during development. The next step, which is the topic of this thesis, is the actual device-side software that will run on physical units in the field. The target platform is the Nordic nRF9151 System-in-Package, which combines an Arm Cortex-M33 application core with an integrated LTE-M and NB-IoT modem on a single chip. The device talks to the GLIDER cloud over MQTT, exchanging JSON-RPC messages for telemetry, configuration, and remote control [1].

In this thesis, we walk through how we designed, built, and tested the firmware for an LTE-M IoT device integrated into the GLIDER platform. We follow a similar software-first approach as in the previous work: the firmware is developed on top of the Zephyr RTOS and the nRF Connect SDK, with most of the application logic exercised first in a native simulator before being flashed onto the nRF9151 development kit. The build is split between a Non-Secure application and the Trusted Firmware-M (TF-M) Secure World running on the same chip, which gives us hardware-backed isolation for keys, secure boot, and firmware updates [2]. Along the way we look at how the device handles peripherals, persists configuration, communicates with the cloud, and what security implications come with leaving such a device unattended on a public mobile network.

Chapter 1

Background and related work

In this chapter we cover the cellular and IoT context our firmware operates in, and we look at how a few comparable products handle the same problem space.

1.1 Cellular IoT and the 2G sunset

1.2 LTE-M and NB-IoT

1.3 The Hardwario GLIDER platform

1.4 Related IoT controllers and firmware

Chapter 2

Hardware platform and software stack

This chapter describes the silicon we target and the software layers we build on. We look at the nRF9151 SiP, the Zephyr RTOS, the nRF Connect SDK and the Trusted Firmware-M build target that the rest of the firmware runs against.

2.1 Nordic nRF9151 SiP

2.1.1 Arm Cortex-M33 and TrustZone-M

2.1.2 Cellular modem and AT interface

2.1.3 Peripherals used by the firmware

2.2 nRF9151 Development Kit

2.3 Zephyr RTOS

2.4 nRF Connect SDK and build targets

2.5 Trusted Firmware-M

2.6 native_sim for host-side validation

Chapter 3

Security considerations for cellular IoT firmware

In this chapter we look at what it means to put a controllable device on a public mobile network. We organise the discussion around three reviewed papers that each cover a different layer: a survey of IoT security as a whole, a focused study of MQTT vulnerabilities, and a survey of the Arm TrustZone hardware foundation we rely on.

3.1 Threat model for a deployed GLIDER device

3.2 IoT security pillars (Sicari et al., 2015)

3.3 MQTT protocol vulnerabilities (Dinculeană & Cheng, 2019)

3.4 Hardware isolation with TrustZone-M (Pinto & Santos, 2019)

3.5 Security roadmap for GLIDER firmware

Chapter 4

Firmware design and implementation

This chapter covers the firmware itself: how we structured the code, how the cellular and application layers fit together, how we handle peripherals, and how we keep state across resets and network outages.

4.1 Architecture overview

4.2 Boot sequence

4.3 LTE connectivity

4.4 MQTT client

4.5 JSON-RPC 2.0 envelope

4.6 Peripheral handling

4.6.1 Analog inputs (SAADC)

4.6.2 GPIO outputs

4.6.3 1-Wire temperature sensors

4.7 Settings persistence

4.8 Automation and calendar

4.9 Offline message queue

4.10 Firmware updates (FOTA)

4.11 Shell interface for development

Chapter 5

Evaluation

In this chapter we walk through how we validated the firmware, what scenarios we exercised, and where the current implementation stands against the original objectives.

5.1 Validation methodology

5.2 Functional verification

5.3 Connectivity and recovery

5.4 Security posture assessment

5.5 Comparison with the existing GLIDER cloud

5.6 Discussion and limitations

Conclusion

Bibliography

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